



Personal

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Languages

Catalan ●●●●●
Spanish ●●●●●
English ●●●●○
German ●●●●○

Skills

Programming

PHP	SQL
Python	HCL
Javascript/TS	HTML/CSS
GO	Bash

Other

AWS	Pyspark
Airflow	Nifi
Docker	Terraform
Postgres	NodeJS
Git	Linux
Circleci	Grafana
Datadog	DBT

AI Skills

- Agentic workflows and loops
- LLM Integration
- System optimization
- AI assisted dev

Soft Skills

- Logical mindset
- Hard working
- Empathic
- Open minded

Bernat Ferrer

Data Engineer - Software Engineer



Professional experience

Data Engineer at Mad Collective

10/2023 - Present Barcelona

- Design, creation and maintenance of data pipelines, processing millions of daily events.
- Management of a data warehouse and a data lake enabling advanced analytics for product teams.
- Infrastructure development using terraform and CI/CD pipeline design with circleci.
- Implemented testing, alerting, and observability solutions to ensure data consistency and transparency.
- Migrated our Redshift cluster to the new generation of nodes with zero downtime.
- Migrated our data engineering observability infrastructure from Datadog to Grafana, documented [here](#)
- Designed and implemented a production AI agent workflow, featuring an autonomous tool-orchestration loop with dynamic context discovery and agent routing.

Software Engineer at Mad Collective

2023 Barcelona

- Developed and maintained a billing-gateway app, integrating our products with payment processors
- Optimized heavy workload use-cases by implementing caching solutions and query refactoring, reducing load time by up to 50 %

Software Engineer at Estimtrack

04/2021 - 12/2022 Barcelona

- Development of a workflow automation tool for hospital employee using hexagonal architecture.
- Implementation of a notification and alerting system using web sockets.
- Designed and implemented an algorithm to unify event identifiers across Outlook and Google Calendar.
- Improved API performance by optimizing database queries and reducing call overhead.

Backend developer at Bekodo

03/2021 - 04/2021 Barcelona

- Developed an app to process and load data from spreadsheets to the cloud, helping small businesses transition to e-commerce.

Data analyst at M-net

09/2018 - 10/2019 Munich

- Analyzed revenue data to identify optimization opportunities and minimize losses.
- Detection and assessment of internal and external fraud.

Physics Researcher at the USM

02/2018 - 12/2019 Munich

- Developed an interface for the visualization of data coming from astrophysical simulations using python.
- Designed and implemented a filter algorithm for detection of light curves.

Education | Certificates

AWS Certified Cloud Practitioner

02/2023

Software Development Master at Assembler Institute

06/2020 - 02/2021 Barcelona

Astrophysics Master at LMU

09/2017 - 09/2019 Munich

Physics degree at UAB

09/2012 - 06/2017 Barcelona

Additionally

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| ■ Teacher at the Migracode non profit organization | present |
| ■ Occasional writer at https://medium.com/@bernatferrerm | present |
| ■ Freelance tech consulting services for tema-litoclean group | 2024/25 |
| ■ Development of a product stock manager for <i>Celler de l'Era</i> . | 2022 |
| ■ Creation and maintenance of a vs-code theme . | 2021 |
| ■ Development of a language learning app in Kotlin | 2020 |
| ■ Internship at the European southern observatory. | 2019 Munich |